

RESEARCH ARTICLE

Automatic Detection and Scoring of Kidney Stones on Noncontrast CT Images Using S.T.O.N.E. Nephrolithometry: Combined Deep Learning and Thresholding Methods

Yingpu Cui,¹ Zhaonan Sun,¹ Shuai Ma,¹ Weipeng Liu,² Xiangpeng Wang,² Xiaodong Zhang,¹ Xiaoying Wang¹ 

¹Department of Radiology, Peking University First Hospital, 8, Xishiku Street, Xicheng District, Beijing, 100034, China

²Beijing Smart Tree Medical Technology Co. Ltd., Beijing, China

Abstract

Purpose: To develop and validate a deep learning and thresholding-based model for automatic kidney stone detection and scoring according to S.T.O.N.E. nephrolithometry.

Procedures: Abdominal noncontrast computed tomography (NCCT) images were retrospectively archived from February 2018 to April 2019 for three parts: a segmentation dataset ($n = 167$), a hydronephrosis classification dataset ($n = 282$), and test dataset ($n = 117$). The model consisted of four steps. First, the 3D U-Nets for kidney and renal sinus segmentation were developed. Second, the deep 3D dual-path networks for hydronephrosis grading were developed. Third, the thresholding methods were used to detect and segment stones in the renal sinus region. The stone size, CT attenuation, and tract length were calculated from the segmented stone region. Fourth, the stone's location was determined. The stone detection performance was estimated with sensitivity and positive predictive value (PPV). The hydronephrosis grading and stone size, tract length, number of involved calyces, and essence grading were estimated with the area under the curve (AUC) method and linear-weighted κ statistics, respectively.

Results: The stone detection algorithm reached a sensitivity of 95.9 % (236/246) and a PPV of 98.7 % (236/239). The hydronephrosis classification algorithm achieved an AUC of 0.97. The scoring model results showed good agreement with radiologist results for the stone size, tract length, number of involved calyces, and essence grading ($\kappa = 0.95$, 95 % confidence interval [CI]: 0.92, 0.98; $\kappa = 0.97$, 95 % CI: 0.95, 1.00; $\kappa = 0.95$, 95 % CI: 0.92, 0.98; and $\kappa = 0.97$, 95 % CI: 0.94, 1.00), respectively.

Conclusions: The scoring model was constructed that can automatically detect and score stones in NCCT images.

Key words: Kidney stone, Multidetector computed tomography, Deep learning, Image processing, Computer-assisted, Severity of illness index

Abbreviations and acronyms: 3D, Three dimensional; CT, Computed tomography; NCCT, Noncontrast computed tomography; PCNL, Percutaneous nephrolithotomy; ROC, Receiver

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Correspondence to: Xiaoying Wang; e-mail: wangxiaoying@bjmu.edu.cn

operating characteristic curve; AUC, Area under the curve; PPV, Positive predictive value; CI, Confidence interval

Introduction

Urolithiasis is the most common disease in urinary surgery and represents a major public health concern with a continuously increasing prevalence both in industrialized nations and developing countries due to changing social and economic conditions [1, 2]. Computed tomography (CT) is the most commonly used imaging modality owing to its high sensitivity and specificity for identifying urinary stones [3].

For the management of large calculi in the upper urinary tract, such as staghorn calculi, percutaneous nephrolithotomy (PCNL) is the current gold-standard method [4]. PCNL has higher stone-free rates (SFRs) and a higher rate of complications than retrograde nephroscopy [5]. As a result, the S.T.O.N.E. nephrolithometry scoring system has been developed and externally validated to enable the appropriate use of PCNL, minimize adverse outcomes, and provide a standardized reporting of stone complexity [6, 7]. Depending on CT imaging, the S.T.O.N.E. score can be easily applied during preoperative planning. This method can generate reproducible results with minimal subjectivity for variables including the stone size (S), tract length (T), obstruction (O), number of involved calyces (N), and essence or stone density (E) (Supplementary Material 1). A structured radiology report of urolithiasis containing these items has been proposed to meet the greater expectations of urologists for stone disease assessment [8].

The patient's S.T.O.N.E. score before PCNL is important for clinical management. Despite its good reproductivity, there is an interobserver variability in the S.T.O.N.E. score system pertaining to the stone size and the number of calyces involved [9]. In addition, it can take considerable amount of time for a surgeon or radiologist to assess every patient due to the high prevalence of urinary stone disease. Automated stone detection and scoring, following standard procedures, could decrease the variability, cost, and surgeon or radiologist burden.

Convolutional neural networks are most commonly used for image processing and have demonstrated good performance in image classification, image segmentation, and object detection [10]. Convolutional neural networks perform well in extracting complicated nonlinear features from images; therefore, they are widely used for complicated work such as kidney segmentation [10]. Other segmentation methods, such as threshold-based methods, are much easier and are used for segmenting and quantifying the size of stones and vascular calcifications [11–14]. However, no prior study has combined convolutional neural networks and threshold-based methods for the automatic detection and

assessment of stones. In addition, there has been a gap between the current models' output and the radiological or clinical assessment of kidney stones.

We hypothesize that a model combining convolutional neural networks and thresholding methods can detect and score kidney stones. The purpose of this study is to develop and validate a scoring model for automatic kidney stone detection and scoring on noncontrast computed tomography (NCCT) images according to the S.T.O.N.E. score system.

Materials and Methods

Dataset

This retrospective study was approved by the institutional review board, and the requirement for informed consent was waived. Figure 1 illustrates the patients' enrollment. At our institution between February 2018 and April 2019, 625 abdominal NCCT images were separately and consecutively collected from a local electronic medical database and were divided into three parts: a dataset for kidney and renal sinus segmentation (segmentation dataset) ($n = 178$), a dataset for hydronephrosis classification (classification dataset) ($n = 314$), and the test set for S.T.O.N.E. grading ($n = 133$). The image acquisition parameters are detailed in Supplementary Material 2. The inclusion criteria for the segmentation dataset were the consecutive NCCT images of February 2018. The inclusion criteria for the classification dataset were the consecutive NCCT images diagnosed as mild to severe hydronephrosis in the reports, either unilateral or bilateral, from March 2018 to December 2018. The inclusion criteria for the test dataset were the consecutive NCCT images containing stones > 2 mm in width in the axial or coronal plane, either solitary or multiple, unilateral or bilateral, from January 2019 to April 2019. In case no large stones were included in the test set for S.T.O.N.E. grading, we extracted CT images diagnosed as staghorn calculus as the test set from March 2018 to April 2019. There was no overlap between the three datasets. The exclusion criteria were as follows: (a) PCNL operation or double-J stent placement ($n = 22$), (b) severe renal anomalies (*e.g.*, renal aplasia, ectopic, horseshoe, and polycystic kidney) ($n = 13$), and (c) radical nephrectomy ($n = 24$).

Image Processing and Labeling

The images were processed with Simple ITK (<http://www.simpleitk.org/>) and were reversed from right to left, aiming to double the cases and focus on the unilateral kidney. The images were labeled kidney and renal sinus by

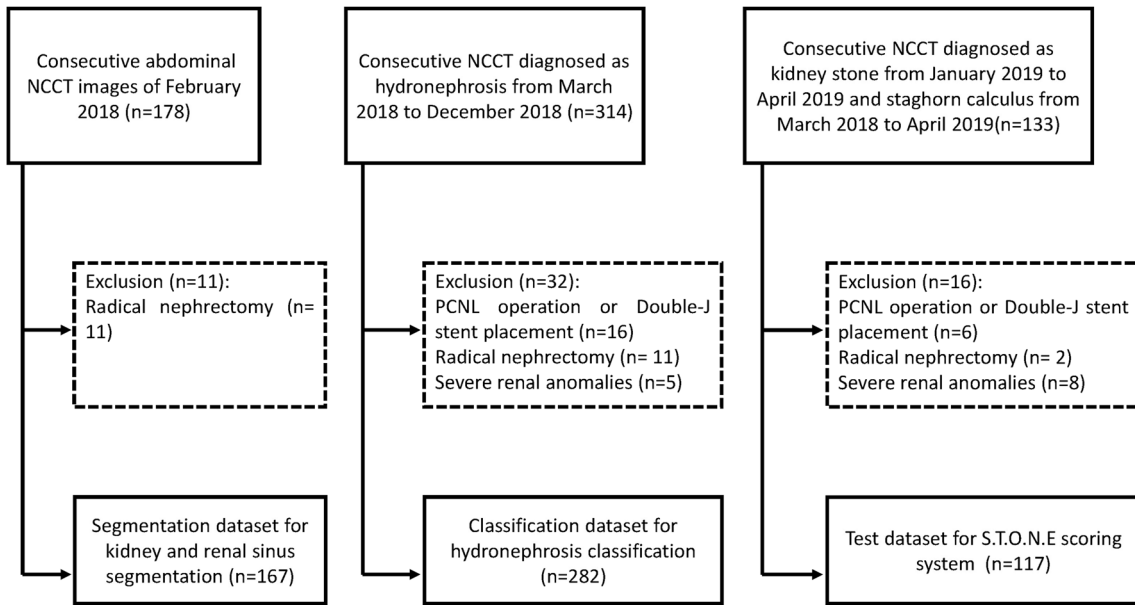


Fig. 1 Diagram shows the patient selection flow. NCCT, noncontrast computed tomography; PCNL, percutaneous nephrolithotomy.

two in-house general radiologists (with 1–2 years of experience) and were validated by a genitourinary radiologist (with more than 15 years of experience). In the classification and test datasets, the CT images were reviewed and scored for hydronephrosis according to the most common system (Society of Fetal Ultrasound, SFU) [15, 16]. We set the binary categorization for grade 0 to 2 as no or mild hydronephrosis and grade 3 to 4 as moderate to severe hydronephrosis by the obstruction degree in the S.T.O.N.E. score system. In the test dataset, the stones with width > 2 mm in the axial or coronal plane were labeled and validated in a bone window (L300/W1120) by the same three radiologists by using a custom semiautomated approach [17, 18]. The stone length, width, size, density, tract length, and volume were derived from manual segmentation (Fig. 2, Supplementary Material 3) [12, 19]. There are three calyceal groups: upper, middle, and lower. The stone location was defined as the upper, middle, lower, upper-middle, middle-lower, and upper-middle-lower calyceal groups (Supplementary Material 3). The stone location was recorded and validated by the same three radiologists. For each kidney, the sum of the stone sizes, the maximal density, and tract length were regarded as the S.T.O.N.E. size, density, and tract length score of this kidney, respectively. The number of involved calyces was also scored. The radiologists were blinded to the model's results.

Modeling

Our scoring model used an ensemble of a segmentation algorithm, a classification algorithm, and several image

processing methods to achieve the automatic scoring of kidney stones. The pipeline is shown in Fig. 2.

Kidney and Renal Sinus Segmentation

We used the 3D U-Net segmentation framework described by Çiçek Ö *et al.* for kidney and renal sinus segmentation [20]. The 3D U-Net architecture can fully exploit volumetric spatial information and thus is a preferred choice for 3D organ segmentation. The schematics for the network architecture are shown in Supplementary Material 4. The kidney and renal sinus segmentation method was based on a cascaded application of two 3D U-Net frameworks. The first 3D U-Net was trained on resampled global images of low resolution to locate kidney through the rough kidney segmentation output. The local kidney images of high resolution were obtained through cropping original images according to the label map predicted by using the first 3D U-Net. The next 3D U-Net was trained on local kidney images of high resolution to segment kidney and renal sinus accurately. To incorporate the sinus border, a morphological dilatation with a spherical structuring element with a 5-pixel radius (5 mm) was applied to the segmented sinus region.

Hydronephrosis Classification

We implemented our classification algorithm by using the same deep 3D dual-path networks described by Zhu WT *et al.* [21]. Supplementary Material 4 contains additional details regarding the classification algorithm.

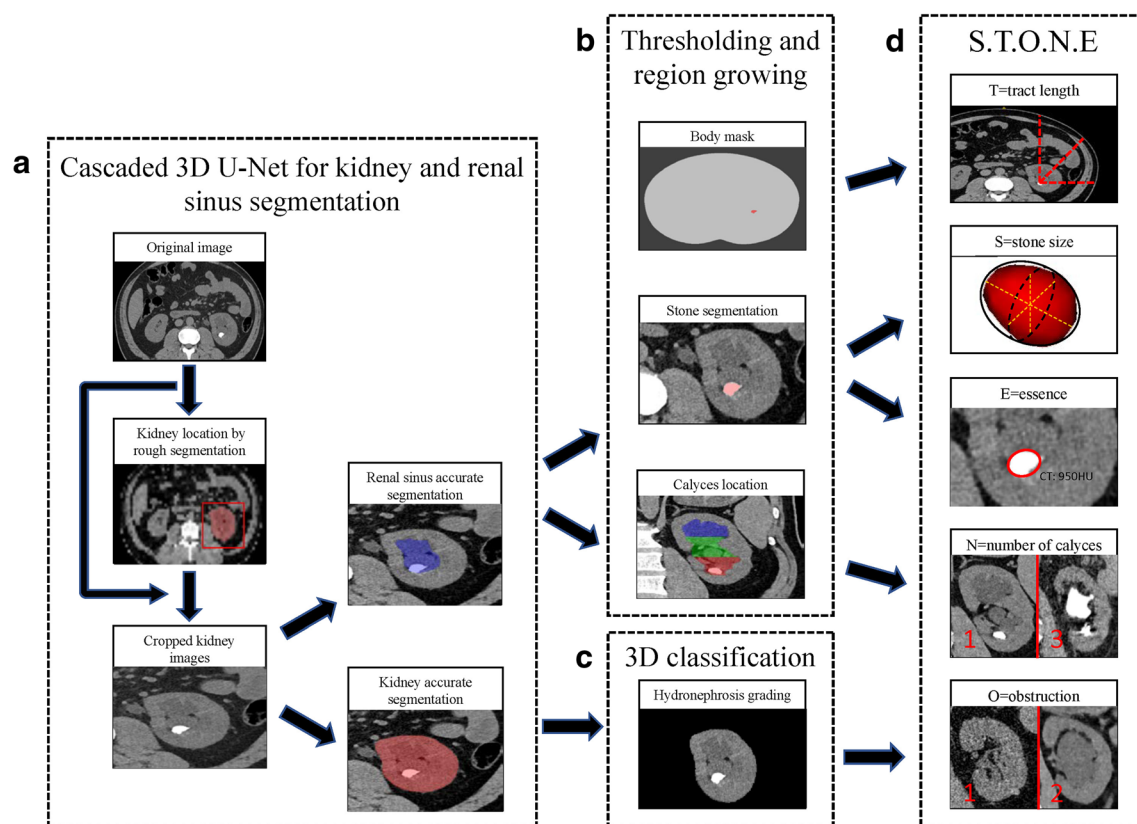


Fig. 2 Workflow for automatic stone scoring according to S.T.O.N.E. nephrolithometry. **A** The cascaded 3D U-Nets for kidney and renal sinus segmentation on noncontrast CT images. The first 3D U-Net was trained on resampled global images of low resolution to locate kidney through the rough kidney segmentation output. The next 3D U-Net was trained on cropped images of high resolution to segment kidney and renal sinus accurately. **B** Imaging processing method for stone detection, segmentation, and location based on the renal sinus region predicted by using the cascaded 3D U-Nets. The body shape was also segmented. **C** The deep 3D dual-path nets for hydronephrosis classification based on the masked kidney images from the cascaded 3D U-Nets. **D** Calculation and output for each component in the S.T.O.N.E. scoring system. The tract length was defined as the mean vertical distance from the center of the stone to the skin measured on a supine NCCT film at 0°, 45°, and 90°. The tract length was calculated using the body's shape and the stone center from the stone region according to the definition of the tract length. The stone size was estimated by combining the measures of the length and width in square millimeters, while the stone length, width, and height were defined as the length along 3 principal directions calculated by principal component analysis for the surface points of the stone. The stone essence was regarded as the maximal attenuation of the segmented stone region. The number of involved calyces was determined by the stone's relative position in the renal sinus. The calix length was defined as each calyceal group's length along the z-axis. The calyceal group was involved when its region contained the stone's center. We assumed that only one calix was involved when the stone length along the z-axis was less than two-thirds the calix length, and full calyces were involved when the length was larger than the calix length. Otherwise, two calyces were involved (Supplementary Material 3). The obstruction assessment was derived from the output of the deep 3D dual-path networks for hydronephrosis classification.

Stone Detection, Segmentation, Measurement, and Tract Length Calculation

Thresholding and region growing were used to detect and segment the stones in the renal sinus region predicted by using the trained 3D U-Nets. The binarization threshold of the thresholding method was set to 250 HU for stone detection [8, 22], and the lower threshold for region growing was defined as one-half of the maximum attenuation value in the stone, with an upper limit of 600 HU and a lower limit of 200 HU [11, 23]. The length, width, size, density, tract length, and volume were

derived from automated segmentation (Fig. 2, Supplementary Material 3). For several stones in a kidney, we regarded the sum of the stone sizes, the maximal density, and tract length of the stones as the stone size, density, and tract length scores, respectively, in the S.T.O.N.E. system for this kidney.

Number of Involved Calyces

We assumed that the renal sinus region acquired from the cascaded 3D U-Nets, divided into three parts equally

along the z-axis, can represent the upper, middle, and lower calyceal groups each. The involved calyces were determined by the stone's relative position of the renal sinus (Fig. 2, Supplementary Material 3).

Model Evaluation and Statistical Analysis

To evaluate the performance of the segmentation network, the Dice coefficient was used [24]. The Dice takes the number of pixels that overlap between the ground truth and automated segmentation and divides them by the total number of pixels assigned to that class by using either method and is widely used for assessment of segmentation. To generalize the reliability of the networks, 5-fold cross-validation was performed (Supplementary Material 4). A total of 80 % of the images were used as training and in-training validation, while the remaining 20 % of the images were used as held-out testing for each fold. The segmentation performance was evaluated on the held-out test dataset each fold and the test set for S.T.O.N.E. grading, respectively.

To assess the performance of the classification network, a receiver operating characteristic (ROC) curve was used, and the sensitivity, specificity, and accuracy were calculated. The classification performance was evaluated by using a 5-fold cross-validation method (Supplementary Material 4).

To assess the performance of the stone detection algorithm, we computed the sensitivity and positive predictive value (PPV) for stone detection. To assess the stone segmentation algorithm, box plots for the distribution of the Dice coefficients were generated.

To assess the performance of the stone measurement algorithm, we computed the absolute difference between the automated and manually derived measurements and displayed the results using Bland–Altman plots. In addition, we computed the agreement between radiologists and the model results according to the grading in the S.T.O.N.E. score system and weighted κ statistics using linear weighting.

To assess the performance of the stone location algorithm, we quantified the types of disagreements in confusion matrices and computed the agreement between radiologists and the model *via* weighted κ statistics using linear weighting.

Continuous variables were summarized as mean $\pm$ standard deviation or median (interquartile range) as appropriate, and categorical variables were summarized using counts (percentage). We computed all data using a programming language (python version 3.5.0; Python Software Foundation; <https://www.python.org>) and statistical software (STATA 15.1, STATA Corporation, College Station, TX).

Results

Patient Characteristics

Among the 625 patients from the three datasets, 566 patients were enrolled (Fig. 1). The segmentation dataset, the classification dataset, and the test dataset consisted of 167 patients (334 kidneys) (mean age, 53.8 ± 18.7 years; range, 13–102 years), 282 patients (564 kidneys) (mean age, 52.5 ± 16.9 years; range, 13–90 years), and 117 patients (234 kidneys) (mean age, 53.7 ± 14.8 years; range, 14–93 years), respectively. In the classification dataset, the radiologists assessed hydronephrosis as no or mild hydronephrosis in 361 of 564 (64.0 %) kidneys and as moderate to severe hydronephrosis in 203 of 564 (36.0 %) kidneys. In the test dataset, the S.T.O.N.E. nephrolithometry results were assessed (Table 1).

Kidney and Renal Sinus Segmentation

The segmentation algorithm consisted of two cascaded 3D U-Nets. For 5-fold cross-validation of 334 kidneys in the segmentation dataset, every fold had 67 or 66 kidneys. The mean cross-validation Dice for the first kidney location procedure was 0.90 ± 0.03 , and the mean cross-validation Dice for the next kidney and renal sinus segmentation procedure was 0.97 ± 0.01 and 0.94 ± 0.01 , respectively (Table 2).

In the test dataset for S.T.O.N.E. scoring, the Dice for the whole kidney and renal sinus segmentation algorithm was 0.97 ± 0.01 and 0.93 ± 0.01 , respectively. We found that all kidneys were well-segmented. In contrast, the renal sinuses of kidneys with severe hydronephrosis and thin cortex were poorly segmented due to the destruction of the anatomy (Fig. 3).

Hydronephrosis Classification

For 5-fold cross-validation of 564 kidneys in the classification dataset, every fold had 113 or 112 kidneys, and the classification algorithm achieved a mean cross-validation testing accuracy, AUC, sensitivity, and specificity of 90.3 %, 0.96, 89.0 %, and 91.2 %, respectively. The detailed discrimination data are presented in Table 3 and Supplementary Material 5.

In the test dataset for S.T.O.N.E. scoring, the AUC for the classification algorithm was 0.97 (95 % confidence interval (CI), 0.94, 0.99). The accuracy, sensitivity, and specificity were 91.9 % (95 % CI, 87.6–95.0), 86.5 % (95 % CI, 74.2–94.4), and 93.4 % (95 % CI, 88.8–96.5), respectively (Table 4). The confusion matrices for obstruction grade in the S.T.O.N.E. system are shown in Fig. 4.

Table 1 S.T.O.N.E. nephrolithometry for the test dataset

S.T.O.N.E. nephrolithometry components	All kidneys ($n = 234$)	Left kidneys ($n = 117$)	Right kidneys ($n = 117$)
O = obstruction			
1: no or mild	182	83	99
2: moderate to severe	52	34	18
S = stone size			
Without stone	103	45	58
1: 0–399 mm ²	102	56	46
2: 400–799 mm ²	8	3	5
3: 800–1599 mm ²	13	7	6
4: > 1599 mm ²	8	6	2
T = tract length			
Without stone	103	45	58
1: ≤ 100 mm	27	13	14
2: > 100 mm	104	57	47
N = number of involved calyces			
Without stone	103	45	58
1: 1 calyx involved	74	43	31
2: 2–3 calyces involved	48	23	25
3: full staghorn calculus	9	6	3
E = essence			
Without stone	103	45	58
1: ≤ 950 HU	46	26	20
2: > 950 HU	85	46	39

Stone Detection, Segmentation, Measurement, and Tract Length Calculation

For stone detection, the algorithm detected and output stones with calculated widths > 2 mm. Of the 266 true stones output by the algorithm, 30 stones were not labeled with a manually measured width ≤ 2 mm by the radiologists. These true stones with manually measured widths < 2 mm were excluded from analysis.

Of the 131 kidneys containing stones and 103 kidneys without stones, 246 true stones were evaluated. A total of 236 true stones and 3 false stones were output by the algorithm. The stone detection algorithm reached a sensitivity of 95.9 % (236/246) and a PPV of 98.7 % (236/239). For the 10 stones of 4 kidneys missed by the algorithm, 8 stones of 2 kidneys were missed owing to severe hydronephrosis and the destruction of the renal sinus. One stone of 1 kidney was missed owing to the lower pole not covered by the renal sinus region of the segmentation algorithm's output. One stone of 1 kidney was not included in the output stones with an automated width ≤ 2 mm. For the 3 mistaken stones, 2 were the calcification of vessels or kidneys and 1 was the suture for partial nephrectomy (Fig. 3).

Regarding stone segmentation, the automated and manual stone segmentation results demonstrated a mean Dice of 0.83. Classified by the stone size grade of the S.T.O.N.E. system, the Dice coefficient distributions were displayed as a box plot (Supplementary Material 6). There are 3 outlines for grades 2, 3, and 4, indicating poorly segmented stones (Fig. 5).

The Bland–Altman plots for stone's length, width, density, and tract length between the automated and manual measurements are shown in Fig. 6. For the stone's length and width, there was a strong tendency toward smaller automatic measurements than manual measurements for the larger stones, which is in accordance with Johan Jendeborg's results [23]. The stone density results, derived from the peak attenuation area, were the same between the automated and manual measurements. The stone size was calculated from the stone length and width, which depended on the accuracy of the stone length and width; we did not perform an extra Bland–Altman test.

For the stone size, density, and tract length grade in the S.T.O.N.E. system, the grade assigned by using the algorithm matched that assigned by the radiologists for 224 of 234 (0.95; 95 % CI: 0.92, 0.98), 230 of 234 (0.97; 95 % CI: 0.95, 1.00), and 229 of 234 (0.97; 95 % CI: 0.94, 1.00)

Table 2 Dice of the segmentation algorithm in the held-out test set for each fold of segmentation dataset

Cross-validation	No. of kidneys	Kidney location	Kidney segmentation	Sinus segmentation
Fold 1	67	0.90	0.97	0.95
Fold 2	67	0.92	0.97	0.94
Fold 3	67	0.86	0.95	0.93
Fold 4	67	0.88	0.97	0.94
Fold 5	66	0.92	0.97	0.93
Mean ± SD	66.8 ± 0.4	0.90 ± 0.03	0.97 ± 0.01	0.94 ± 0.01

SD, standard deviation

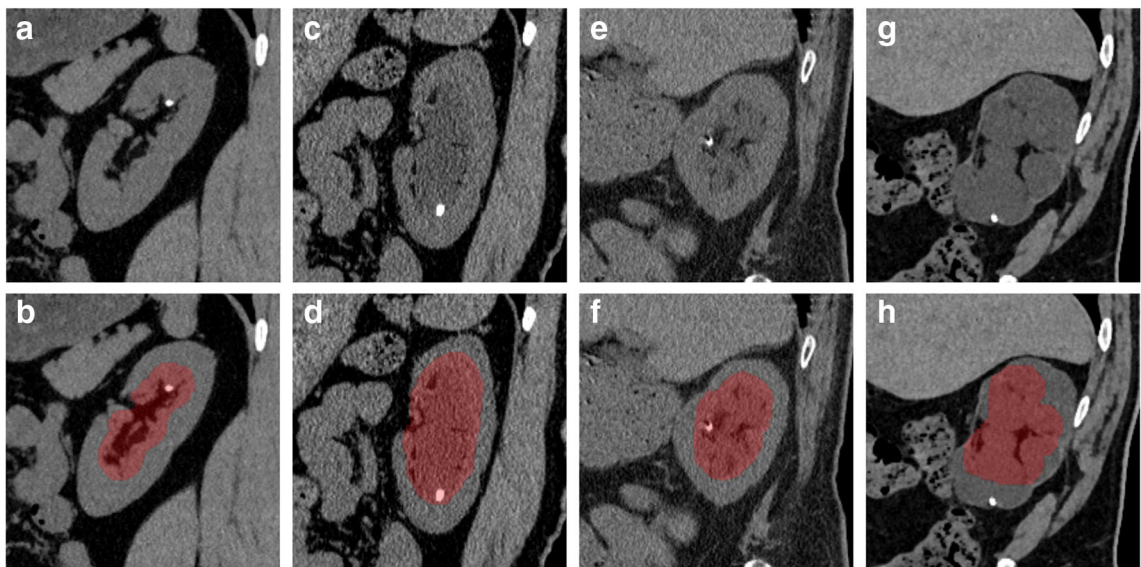


Fig. 3 Images show the examples of 3D U-Net segmentation with morphological dilatation of renal sinus. **a, b** A 45-year-old male with a kidney stone and no hydronephrosis; the sinus was well-segmented and the stone was detected. **c, d** A 48-year-old female with a kidney stone and severe hydronephrosis; the sinus was well-segmented and the stone was detected. **e, f** A false-positive stone in a 74-year-old male; the sinus was well-segmented and the vessel calcification was mistaken as kidney stone. **g, h** A false-negative stone in a 55-year-old male; the sinus was poorly segmented due to severe hydronephrosis and destruction of normal anatomy. The stone was missed for not covered by the renal sinus region.

kidneys, respectively (Table 4). The model downgraded 2 of the 234 (0.9 %) kidneys to grade 0 for the false-negative stones and upgraded 2 of the 234 (0.9 %) kidneys from grade 0 for the false-positive stones (Fig. 4).

Number of Involved Calyces

In the test dataset, the confusion matrices for the assessment of the stone location algorithm are shown in Supplementary Material 7. Agreement between the algorithm and radiologists was good ($\kappa=0.93$; 95 % CI: 0.89, 0.96). With respect to the grade of the number of involved calyces in the S.T.O.N.E. system, the grade assigned by using the algorithm matched that assigned by the radiologists for 224 of 234 ($\kappa=0.95$; 95 % CI: 0.92, 0.98) kidneys (Table 4). The model downgraded 2 of the 234 (0.9 %) kidneys to grade 0 for the false-negative stones and upgraded 2 of the 234 (0.9 %) kidneys from grade 0 for the false-positive stones (Fig. 4).

Discussion

Scoring is important for the management of kidney stones in stone complexity grading and for predicting outcomes after PCNL. In this study, we attempted to develop and validate an automated pipeline for the assessment of kidney stones according to the S.T.O.N.E. system, combining convolutional neural networks and several image processing methods. Based on NCCT images, the model can be easily applied in clinical practice despite its complex algorithms. During the procedure, no steps require a urologist or radiologist, thus generating reproducible results with no subjectivity and enabling a low-cost and low-time-consuming assessment of kidney stones. For the test dataset, our model achieves good agreement between the algorithms and radiologists. The stone volume has also demonstrated prognostic value for predicting clinically relevant outcomes and recurrent stone events after percutaneous nephrolithotomy [24]. Our model derived stone volume

Table 3 Discrimination of the classification algorithm in the held-out test set for each fold of classification dataset

Cross-validation	No. of kidneys	AUC	Sensitivity (%)	Specificity (%)	Accuracy (%)
Fold 1	113	0.96 (0.91, 0.99)	86.4 (72.6, 94.8)	91.3 (82.0, 96.7)	89.4 (82.2, 94.4)
Fold 2	113	0.97 (0.92, 1.00)	89.5 (75.2, 97.1)	92.0 (83.4, 97.0)	91.2 (84.3, 95.7)
Fold 3	113	0.95 (0.89, 0.98)	85.1 (71.7, 93.8)	95.5 (87.3, 99.1)	91.2 (84.3, 95.7)
Fold 4	113	0.97 (0.92, 0.99)	93.9 (79.8, 99.3)	91.3 (82.8, 96.4)	92.0 (85.4, 96.3)
Fold 5	112	0.97 (0.91, 0.99)	90.2 (76.9, 97.3)	85.9 (75.6, 93.0)	87.5 (79.9, 93.0)
Mean $\pm$ SD	112.8 $\pm$ 0.4	0.96 $\pm$ 0.01	89.0 $\pm$ 3.5	91.2 $\pm$ 3.4	90.3 $\pm$ 1.8

Data in parentheses are 95 % confidence intervals
AUC, area under the curve; SD, standard deviation

Table 4 Comparison of radiologists and the scoring model assessment for the S.T.O.N.E. scoring system in the kidney level

S.T.O.N.E. components	Accuracy	No. of kidneys
Obstruction	91.9 % (87.6, 95.0)	234
S.T.O.N.E. components	Linear-weighted κ value	No. of kidneys
Stone size	0.95 (0.92, 0.98)	234
Tract length	0.97 (0.95, 1.00)	234
Number of involved calyces	0.95 (0.92, 0.98)	234
Essence	0.97 (0.94, 1.00)	234

Data in parentheses are 95 % confidence intervals
No., number

measurements from stone segmentation, which is a more precise description than the stone size.

Several steps in our workflow have been explored by other groups. In terms of kidney and renal sinus segmentation, Oliveira *et al.* proposed a method using a coupled formulation of the B-spline explicit active surfaces (BEAS) framework to segment a kidney and renal sinus [25]; however, the Dice for the renal sinus was not satisfactory. For kidney anatomical segmentation, Papalia *et al.* proposed a methodology for reporting kidney tumor location by dividing a kidney into 12 standard segments [26]; however, the methodology for the location of the calyces was not mentioned. For stone detection, Langkvist *et al.* developed convolutional neural networks to detect ureteral stones and proposed a probabilistic approach to

include anatomical information [27]. Instead of segmenting the ureter in the first step, the method is less plausible for medical diagnosis. In contrast, for stone detection, our model follows the disease diagnosis pipeline by integrating medical knowledge into model development. To ascertain the involved calyces and eliminate the impact of the calcifications of kidneys, we detected and located stones in the renal sinus region instead of the renal region.

There are several limitations to our study. First, the images used for model development and testing were acquired at a single radiology department, and more images from different institutions would be needed to improve the generalizability. Second, considering that S.T.O.N.E. is used for preoperative assessment, kidneys after PCNL operation or double-J stent placement were excluded from our study. Kidneys with severe anomalies were also excluded from our study, and a more robust model to segment such kidneys is needed. Third, the renal sinus region instead of the collecting system was segmented owing to the complexity and variability of the collecting system. As a result, one vascular calcification was mistaken as a renal stone. However, no model for renal collecting system segmentation in NCCT has been proposed. Fourth, although our model reached good agreement with radiologists for stone location, the calyces were simply segmented by dividing the renal sinus

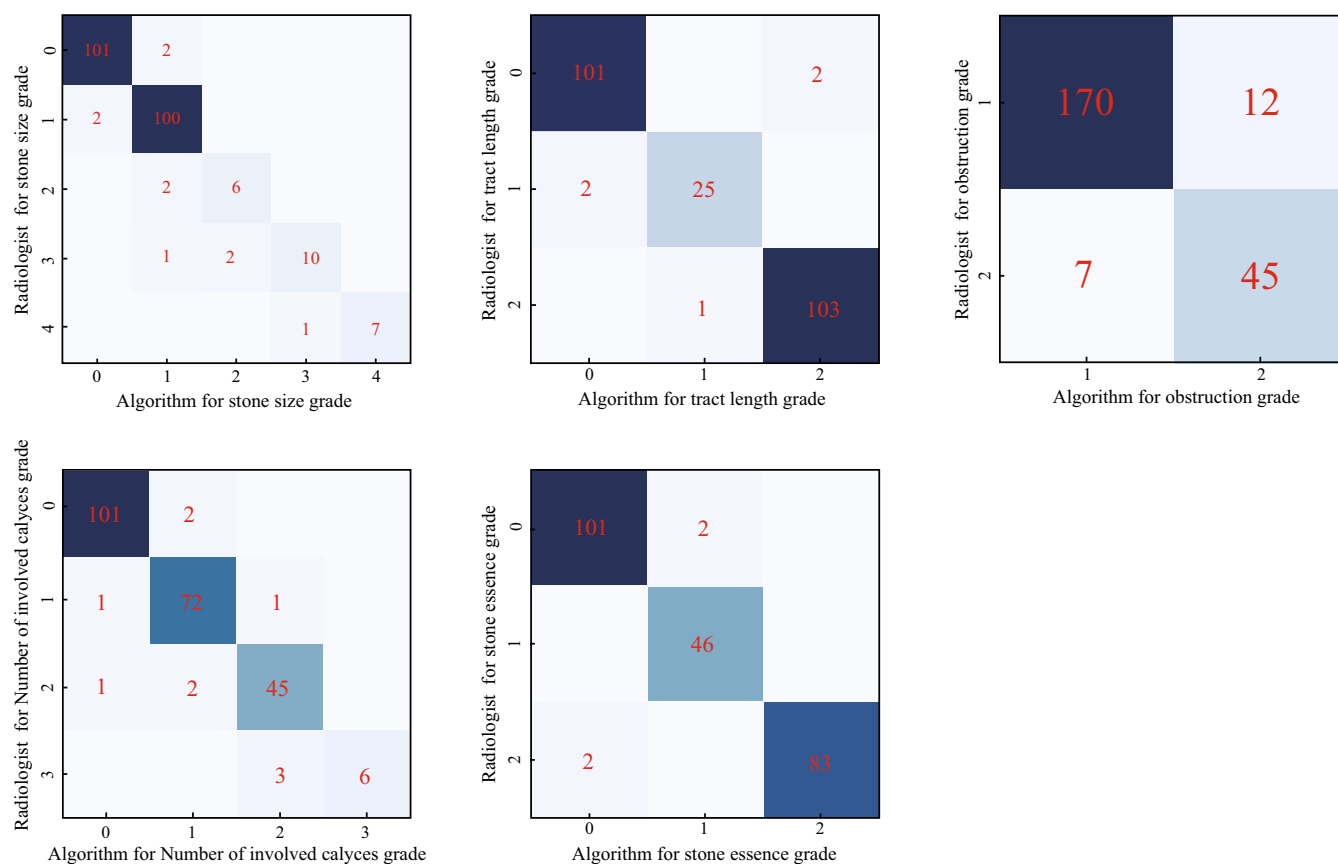


Fig. 4 Comparison of radiologists and the algorithm assessment for the S.T.O.N.E. scoring system in the kidney level. Grade 0 indicates no stone in the kidney.

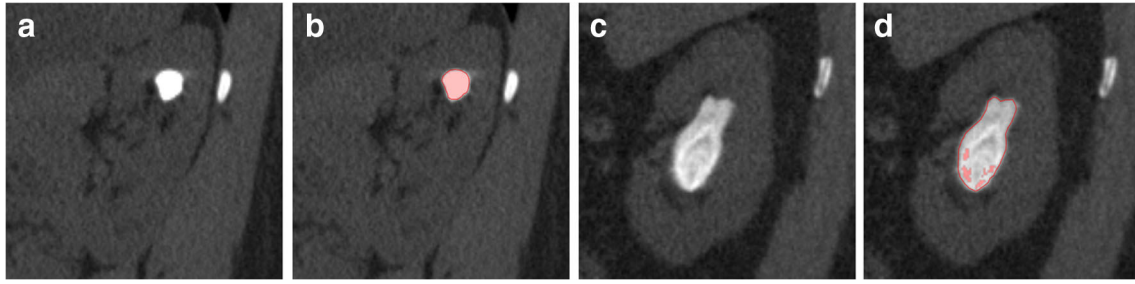


Fig. 5 Images show the examples of raw image and stone segmentation based on threshold segmentation. **a, b** A 45-year-old female with a well-segmented kidney stone. **c, d** A 64-year-old male with a poorly segmented kidney stone. The red line indicates the manual segmentation border of the stone, while the red-painted region indicates automatic stone segmentation.

region equally into three parts, regardless of its anatomy and variability. In the future, we plan to segment and depict the collecting system more accurately. Fifth, for the stone measurements, we noticed that the automatic output seems to be smaller than the manual measurement for larger stones,

which is in accordance with Johan Jendeborg's results [23]. Segmenting stones with the variable threshold method may be suitable for pure stones composed of a single composition [11, 12]. However, it may yield different results for stones of

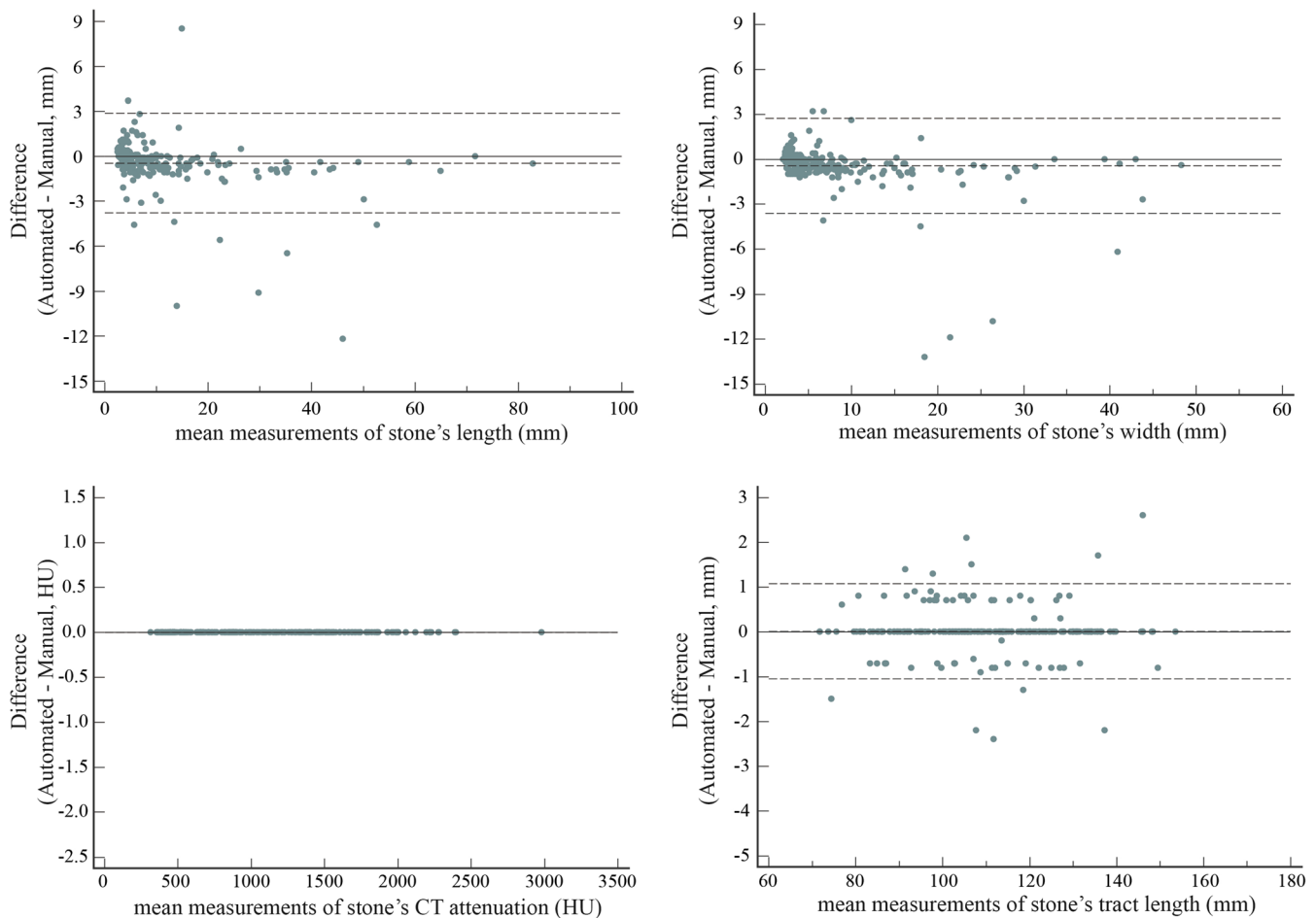


Fig. 6 Automated stone segmentation and measurement results. Bland-Altman plot comparing automated and manual measurements for stone's length, width, CT attenuation, and tract length from 236 stones detected by using the algorithm. The Bland-Altman 95 % limits of agreement between the automated algorithm and the manual measurements were -0.4 ± 3.4 mm, -0.4 ± 3.2 mm, 0.0 ± 0.0 HU, and 0.0 ± 1.1 mm for the stone length, width, CT attenuation, and tract length. Dashed lines delineate the mean and 95 % of stones as calculated by differences between automated and manual measurements. The solid black line indicates the zero line.

mixed composition, as illustrated by 3 poorly segmented large stones in our study, which need further research.

Conclusions

In this study, we developed and validated the deep learning– and thresholding-based model for automatic kidney stone scoring using the S.T.O.N.E. system. The model results showed good agreement with radiologist results. Automatic and standard stone assessment is important for increasing the usage of the stone score system, thus enabling the appropriate use of PCNL and minimizing adverse outcomes. The automatic anatomical segmentation of kidney and kidney stones can enable more detailed information regarding the anatomical features, calyceal distribution, and stone characteristics, such as the stone volume and stone density distribution, and hopefully can provide more factors for evaluating kidney stones. Despite the promise of this application, our methods are not suitable for independent use in a clinical setting until the limitations mentioned above are solved.

Compliance with Ethical Standards

Conflict of Interest

The authors declare that they have no conflict of interest.

Ethical Approval

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments. For this type of study, formal consent is not required.

References

- Park S (2007) Medical management of urinary stone disease. *Expert Opin Pharmacother* 8:1117–1125
- Scales CD, Tasian GE, Schwaderer AL, Goldfarb DS, Star RA, Kirkali Z (2016) Urinary stone disease: advancing knowledge, patient care, and population health. *Clin J Am Soc Nephrol* 11:1305–1312
- Allison SJ (2014) Stones: ultrasonography and computed tomography: performance in detection of kidney stones. *Nat Rev Nephrol* 10:611
- Turk C, Petrik A, Sarica K et al (2016) EAU guidelines on interventional treatment for urolithiasis. *Eur Urol* 69:475–482
- De S, Autorino R, Kim FJ et al (2016) Corrigendum re: “Percutaneous nephrolithotomy versus retrograde intrarenal surgery: a systematic review and meta-analysis” [Eur Urol 2015;67:125–37]. *Eur Urol* 69:e85
- Okhunov Z, Friedlander JJ, George AK, Duty BD, Moreira DM, Srinivasan AK, Hillelsohn J, Smith AD, Okeke Z (2013) S.T.O.N.E. nephrolithometry: novel surgical classification system for kidney calculi. *Urology* 81:1154–1159
- Noureddin YA, Elkoushy MA, Andonian S (2015) External validation of the S.T.O.N.E. nephrolithometry scoring system. *Can Urol Assoc J* 9:190–195
- Kambadakone AR, Eisner BH, Catalano OA, Sahani DV (2010) New and evolving concepts in the imaging and management of urolithiasis: urologists’ perspective. *Radiographics* 30:603–623
- Okhunov Z, Helmy M, Perez-Lansac A, Menhadjji A, Bucur P, Kolla SB, Cho JS, Osann K, Lusch A, Landman J (2013) Interobserver reliability and reproducibility of s.T.o.N.e. nephrolithometry for renal calculi. *J Endourol* 27:1303–1306
- Chartrand G, Cheng PM, Vorontsov E, Drozdal M, Turcotte S, Pal CJ, Kadoury S, Tang A (2017) Deep learning: a primer for radiologists. *Radiographics* 37:2113–2131
- Demehri S, Kalra MK, Rybicki FJ, Steigner ML, Lang MJ, Houseman EA, Curhan GC, Silverman SG (2011) Quantification of urinary stone volume: attenuation threshold-based CT method—a technical note. *Radiology* 258:915–922
- Duan X, Wang J, Qu M, Leng S, Liu Y, Krambeck A, McCollough C (2012) Kidney stone volume estimation from computerized tomography images using a model based method of correcting for the point spread function. *J Urol* 188:989–995
- Hoffmann U, Kwait DC, Handwerker J, Chan R, Lamuraglia G, Brady TJ (2003) Vascular calcification in ex vivo carotid specimens: precision and accuracy of measurements with multi-detector row CT. *Radiology* 229:375–381
- Jacobs C, van Rikxoort EM, Scholten ET, de Jong PA, Prokop M, Schaefer-Prokop C, van Ginneken B (2015) Solid, part-solid, or non-solid?: classification of pulmonary nodules in low-dose chest computed tomography by a computer-aided diagnosis system. *Investig Radiol* 50:168–173
- Fernbach SK, Maizels M, Conway JJ (1993) Ultrasound grading of hydronephrosis-introduction to the system used by the society-for-fetal-urology. *Pediatr Radiol* 23:478–480
- Maizels M, Keays M, Snyder H, Leonard M (2008) Reliability assessment of Society for Fetal Urology ultrasound grading system for hydronephrosis-discussion. *J Urol* 180:1683–1683
- Eisner BH, Kambadakone A, Monga M, Anderson JK, Thoreson AA, Lee H, Dretler SP, Sahani DV (2009) Computerized tomography magnified bone windows are superior to standard soft tissue windows for accurate measurement of stone size: an in vitro and clinical study. *J Urol* 181:1710–1715
- Yushkevich PA, Piven J, Hazlett HC, Smith RG, Ho S, Gee JC, Gerig G (2006) User-guided 3D active contour segmentation of anatomical structures: significantly improved efficiency and reliability. *Neuroimage* 31:1116–1128
- Ward AD, Hamarneh G, Ashry R, Schweitzer ME (2007) 3D shape analysis of the supraspinatus muscle: a clinical study of the relationship between shape and pathology. *Acad Radiol* 14:1229–1241
- Çiçek Ö, Abdulkadir A, Lienkamp SS, Brox T, Ronneberger O (2016) 3D U-Net: learning dense volumetric segmentation from sparse annotation. *Springer International Publishing, Cham*, pp. 424–432
- Zhu WT, Liu CC, Fan W, Xie XH (2018) DeepLung: deep 3D dual path nets for automated pulmonary nodule detection and classification. 2018 IEEE Winter Conference on Applications of Computer Vision (Wacv 2018) 2018:673–681
- Ziamba JB, Li P, Gumani R, Kawamoto S, Fishman EK, Fung G, Ludwig WW, Stoianovici D, Matlaga BR (2018) A user-friendly application to automate CT renal stone measurement. *J Endourol* 32:685–691
- Jendeberg J, Geijer H, Alshamari M, Liden M (2018) Prediction of spontaneous ureteral stone passage: automated 3D-measurements perform equal to radiologists, and linear measurements equal to volumetric. *Eur Radiol* 28:2474–2483
- Selby MG, Vrtiska TJ, Krambeck AE, McCollough CH, Elsherbiny HE, Bergstralh EJ, Lieske JC, Rule AD (2015) Quantification of asymptomatic kidney stone burden by computed tomography for predicting future symptomatic stone events. *Urology* 85:45–50
- Oliveira B, Torres HR, Queiros S, Morais P, Vilaca JL (2018) Segmentation of kidney and renal collecting system on 3D computed tomography images 2018 IEEE 6th International Conference on Serious Games and Applications for Health (SeGAH),
- Papalia R, Abreu ALD, Panebianco V et al (2015) Novel kidney segmentation system to describe tumour location for nephron-sparing surgery. *World J Urol* 33:865–871
- Langkvist M, Jendeberg J, Thunberg P, Loutfi A, Liden M (2018) Computer aided detection of ureteral stones in thin slice computed tomography volumes using convolutional neural networks. *Comput Biol Med* 97:153–160